JUNCHENG LU

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EDUCATION

Teachers College, Columbia University *September 2025 – Present*
M.A. in Cognitive Science in Education
GPA: 3.90/4.00
Selected Coursework: Computational Cognitive Neuroscience; Brain Imaging Data Modeling; fMRI: Principles, Experiment Design and Data Analysis

Beijing Normal University *September 2020 – July 2024*
B.S. in Psychology
GPA: 3.45/4.00

RESEARCH EXPERIENCE

Research Assistant, Social Cognitive and Neural Sciences Lab *September 2025 – Present*
PI: Dr. Jon Freeman *New York, NY*

- Lead computational investigations of social decision-making and metacognitive insight into implicit bias using large-scale Project Implicit behavioral data.
- Develop model-based analysis pipelines in R and Python using MPT/Quad models, hierarchical Bayesian estimation, drift-diffusion models, and mixed-effects regression.
- Integrate latent cognitive parameters with self-report and neuroimaging measures to test mechanistic accounts of bias, threat processing, and cognitive control.

Research Assistant, Intergroup Relation Neuroscience Lab *March 2022 – June 2025*
PI: Dr. Xiaochun Han *Beijing, China*

- Contributed to the design and execution of behavioral and EEG studies of race-dependent collateral-harm decisions during intergroup conflict.
- Applied hierarchical Bayesian DDMs to choices and response times and analyzed P2/N2 ERP components to link latent decision caution with neural conflict monitoring.
- Contributed to data collection, analysis, interpretation, visualization, and manuscript revision for a revised submission currently under review at JESP.

MANUSCRIPTS

Under Review

Lu, J.*, Zhang, Y.*, Han, X., Xiao, Z., & Han, X.† (2026). Computational and neural mechanisms of race-dependent influence of civilian suffering on collateral-harm decisions during intergroup conflicts. *Revised manuscript under review at Journal of Experimental Social Psychology.*

In Preparation Zhang, Y.*, **Lu, J.***, & Han, X.† (2026). Cognitive mechanism of killing-decision discrimination between combatants and civilians during intergroup conflict: The influence of social-role information. *Manuscript in preparation.* **Lu, J.**, Klein, S., & Freeman, J.† (2026). Ingroup-affirming accuracy asymmetry in bias prediction. *Manuscript in preparation.*

*Equal contribution. †Corresponding author.

CONFERENCE PRESENTATIONS

Lu, J., & Han, X.† (2024, April). Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. Accepted for oral and poster presentation at the *Social and Affective Neuroscience Society Annual Meeting*, Toronto, Canada. (Not presented due to visa restrictions.)

SELECTED RESEARCH PROJECTS

Metacognitive Accuracy for Ingroup- and Outgroup-Affirming Implicit Bias *January 2026 – Present*
Independent project, Social Cognitive and Neural Sciences Lab

- Analyze 2,986 participant-by-IAT observations from 1,866 participants across four race and age IATs to test status-based asymmetries in awareness of implicit bias.
- Fit individual-level Quad models by maximum likelihood and hierarchical QPMs in TreeBUGS; use complementary HDDMs to characterize association activation, detection, guessing, bias overcoming, and decision dynamics.
- Relate latent parameters to directional and continuous prediction accuracy using logistic and linear mixed models, explicit-attitude covariates, nested model comparisons, and convergence diagnostics.

Computational Modeling of the Weapon Identification Task *September 2025 – December 2025*
Independent project, Social Cognitive and Neural Sciences Lab

- Applied DDM and MPT models to decompose biased weapon-identification decisions into latent evidence-accumulation and processing components.
- Compared competing cognitive architectures and integrated model parameters with fMRI activation measures to test accounts of threat processing and cognitive control.

Computational and Neural Mechanisms of Collateral-Harm Decisions *September 2022 – June 2025*
Undergraduate thesis, Intergroup Relation Neuroscience Lab

- Examined two studies ($N = 983$) combining a cross-cultural behavioral experiment with EEG to test how civilian race and pain expressions shape collateral-harm decisions.
- Identified a latent Race $\times$ Expression effect on DDM boundary separation despite null effects in model-free behavior; linked race-dependent P2/N2 responses to decision caution.

TECHNICAL SKILLS

Programming	Python (HDDM, NumPy, pandas), R (brms, lme4, TreeBUGS, rtmpt), MATLAB (EEGLAB, SPM12, Psychtoolbox), HTML/JavaScript (jsPsych)
Modeling & Statistics	Hierarchical Bayesian modeling, drift-diffusion models, MPT/Quad models, GLMM/LMM, MCMC diagnostics, model comparison
Neuroimaging	EEG/ERP preprocessing and analysis (EEGLAB), fMRI preprocessing and modeling (SPM12), model-based brain–behavior analysis